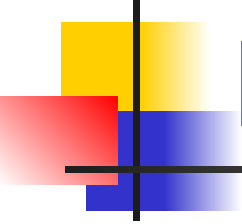


RENEWABLE ENERGY

Renewable Energy

Resource - An essentially inexhaustible energy resource on a human time scale.





PASSIVE SOLAR HEATING

Passive solar heating – captures sunlight directly with a structure and converts it to low-temperature heat for space heating.

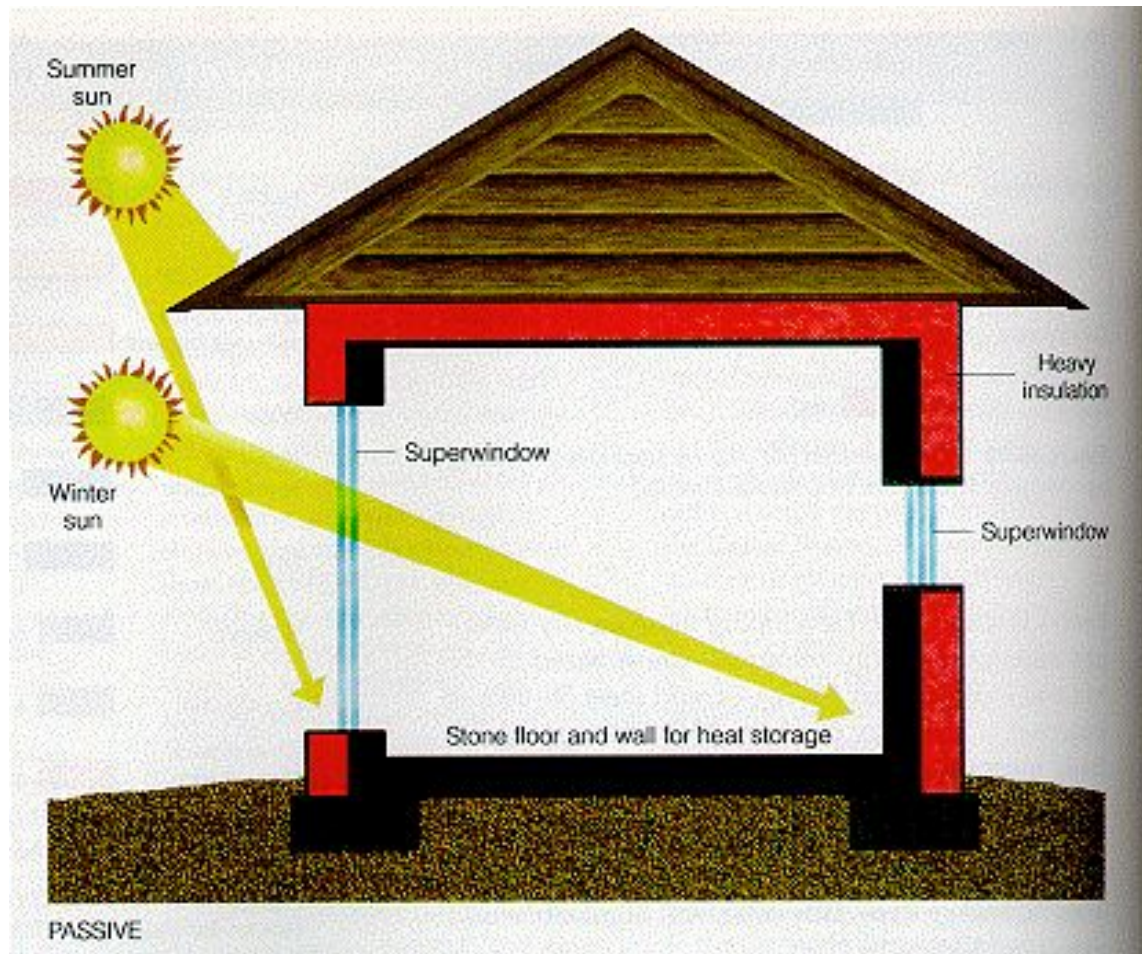
Advantages

1. save money
2. create 2-5 more jobs/unit of electricity
3. eliminate/reduce fossil fuels
4. less pollution
5. less environmental damage

Disadvantages

1. expensive for initial costs
2. aesthetically not pleasing
3. latitude

PASSIVE SOLAR HEATING





ACTIVE SOLAR HEATING

Active solar heating – specially designed collectors absorb solar energy and fan/pump distributes energy to parts of a building to meet space/water heating needs.

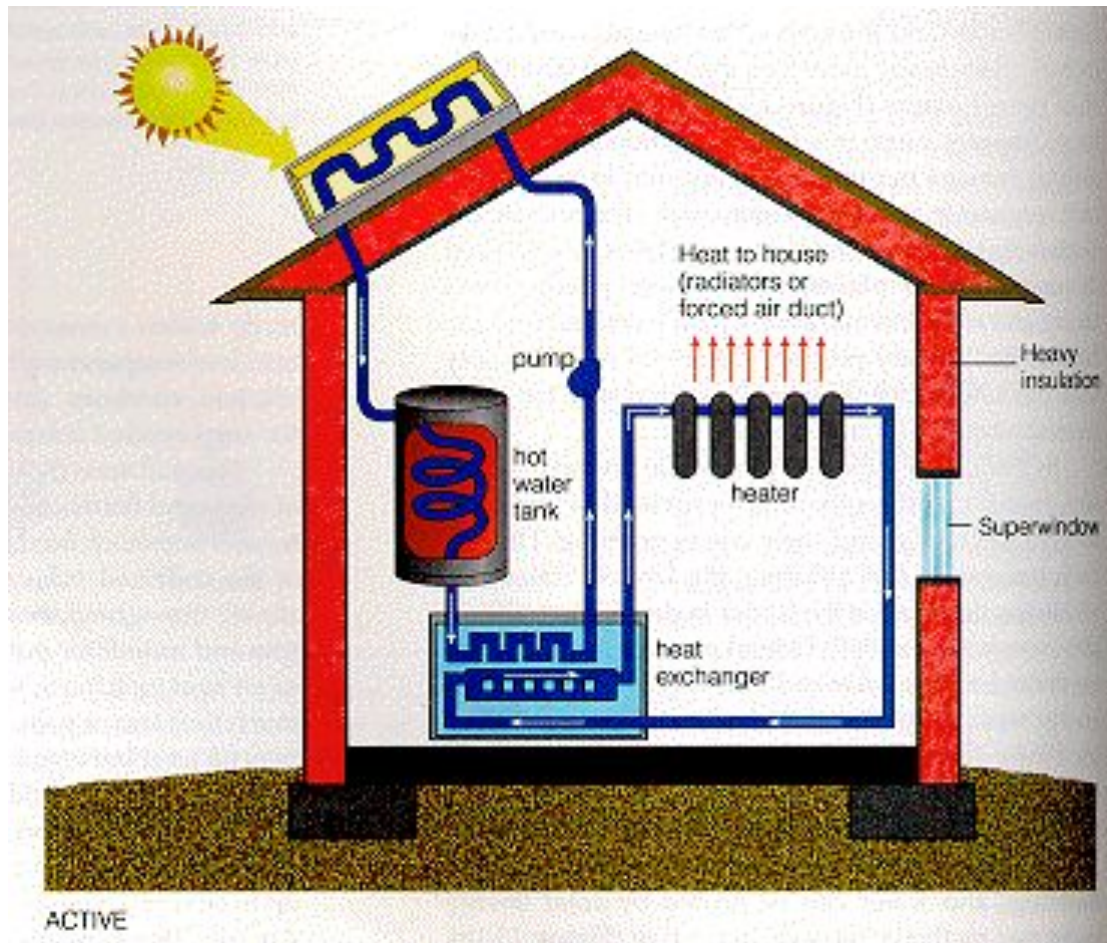
Advantages

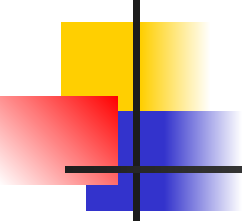
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Disadvantages

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3. latitude

ACTIVE SOLAR HEATING





SOLAR POWER TOWER

Solar Power Tower—huge arrays of computer controlled mirrors that track the sun and focus sunlight on a central heat collection tower.
(Mojave desert in California)

Advantages

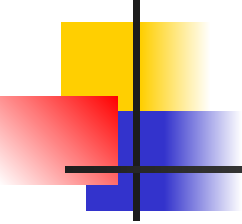
Cost will drop as
Technology improves

Disadvantages

Costs 8X more to
build

SOLAR POWER TOWER





SOLAR THERMAL PLANT

1 Solar Thermal Plant – sunlight is collected and focused on oil-filled pipes that run through the middle of curved solar collectors.

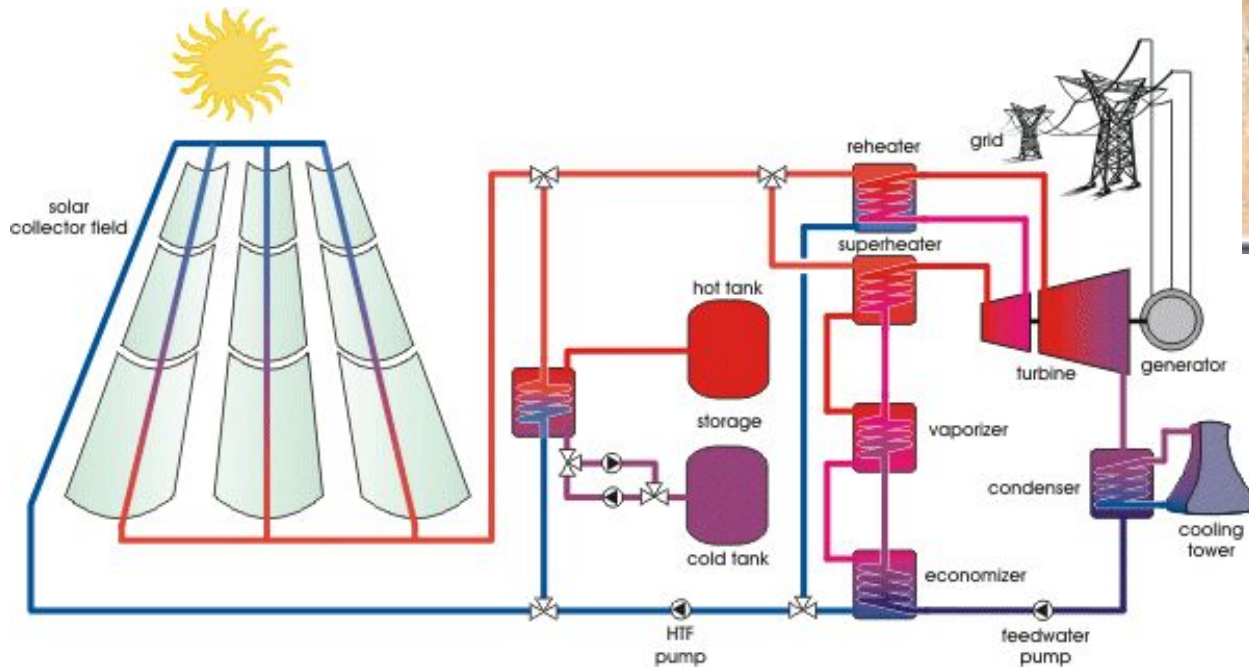
Advantages

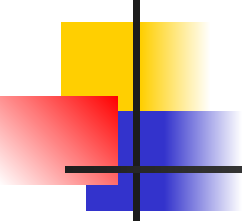
- _1. Can generate temperatures high enough for industrial processes
- 2. can supply back-up electricity
- 3. cheaper than nuclear

Disadvantages

- 1. central receivers are expensive to operate.

SOLAR THERMAL PLANT





SOLAR COOKER

1 **Solar Cooker** – focuses and concentrates sunlight in a box typically covered in glass to trap infrared radiation waves to cook food in rural villages in developing countries.

Advantages

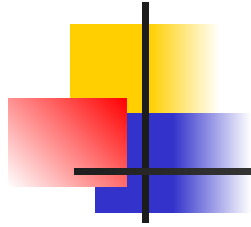
Does not reduce deforestation

Disadvantages

2-4 hours to cook average meal.

SOLAR COOKER





SOLAR HYDROGEN

Solar-Hydrogen – Water can be split into gaseous hydrogen and oxygen.

It is in its infancy. So far... we can create fuel cells where hydrogen and oxygen combine to produce an electrical current, but it is difficult to store enough hydrogen gas in a fuel tank for very long.

□ Politics and economics are “holding up” this technology.

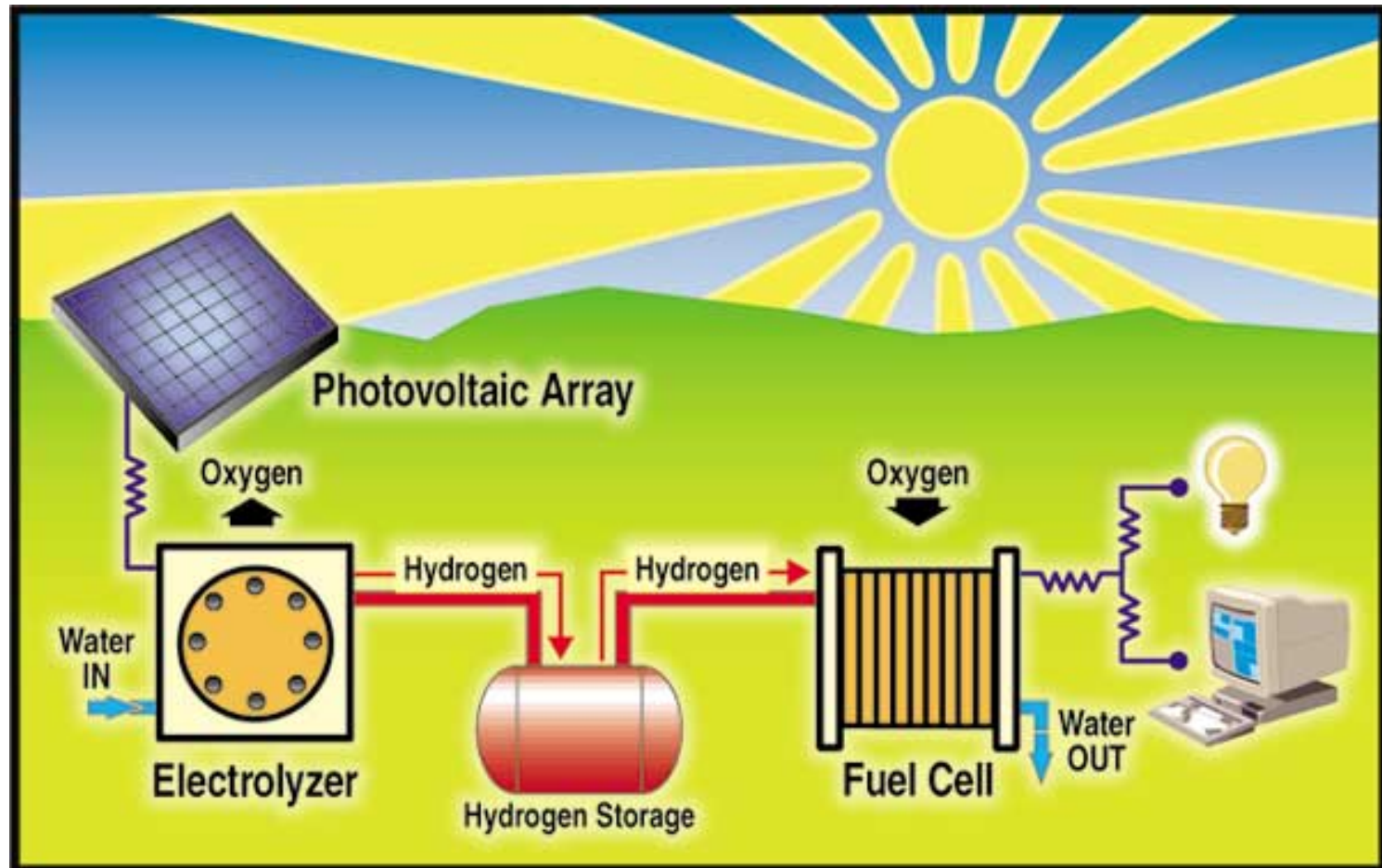
□ R&D from government needed.

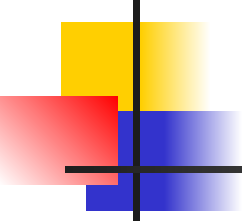
□ must convince energy companies and investors to \$ into this type of power and phase out fossil fuels.

□ must convince public to change over.

...Not Yet!

SOLAR HYDROGEN





HYDROPOWER

1. **Hydroelectric power plants** – A dam is built across a large river to create a reservoir. The higher the head, the greater the amount of power that can be generated. Water is stored in a reservoir during low electricity production. Water is released and flows are controlled as electricity demands peak. Water spins the turbines in the “powerhouse”. Electricity is distributed to end user.

Examples – Aswan High Dam (Egypt) and Colorado River Basin (USA/Mexico)

Advantages

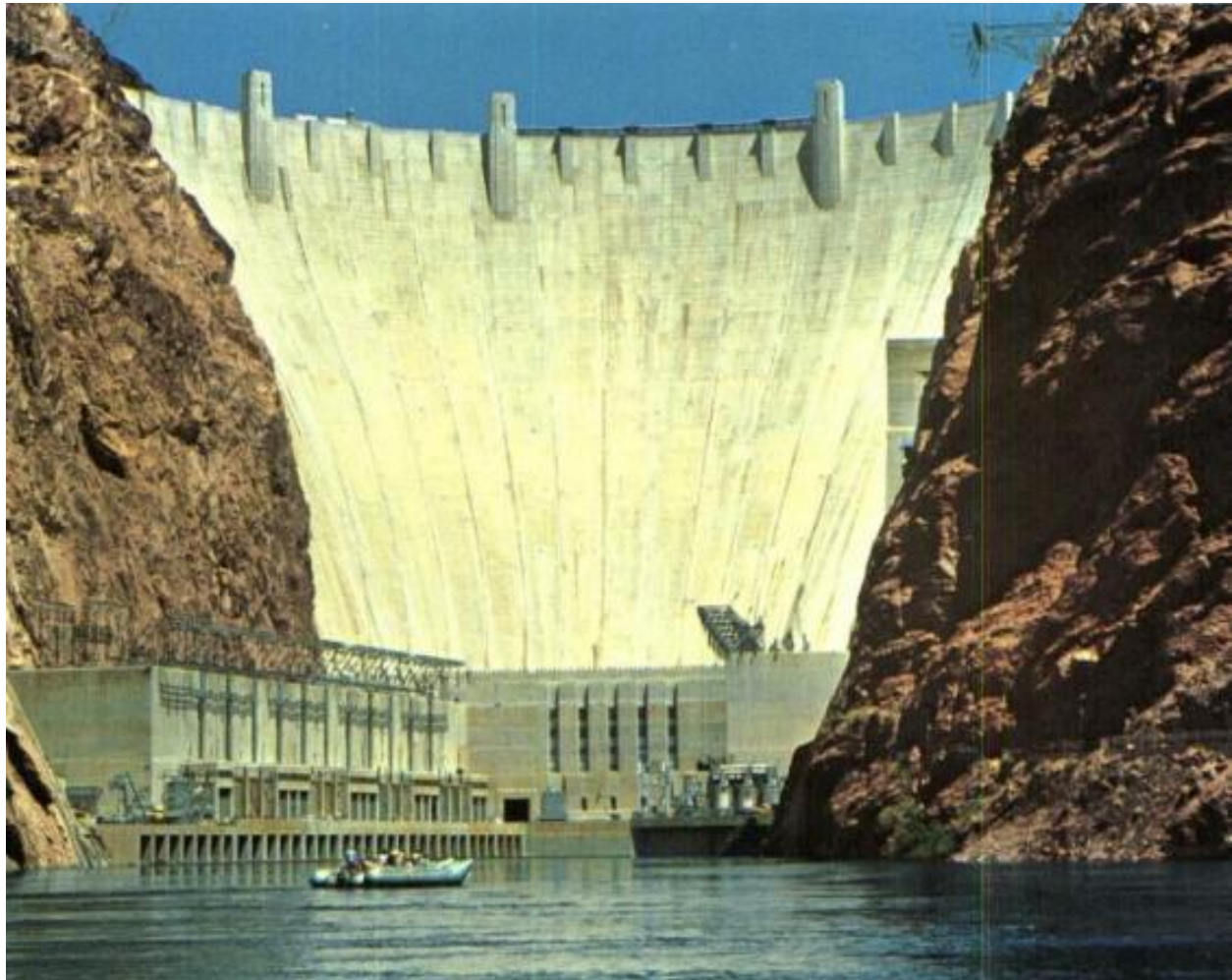
1. Moderate to high energy yield
2. low operating/maintenance costs
3. low air pollution
4. 2-10 times longer life than other pesticides/algicides used
5. Power sources

Disadvantages

1. create floods
2. destroys habitats
3. uproots people
4. 2-10 x longer lifespan than other
5. Decreases fish harvests

HYDROPOWER

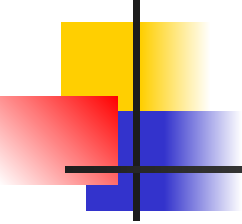
Hoover
Dam



HYDROPOWER

Aswan
High
Dam





TIDAL POWER

1Tidal Power- power created from tidal energy

Advantages

1. tidal energy spins turbines

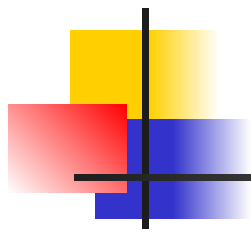
Disadvantages

1. few suitable sites

2. Construction costs high

TIDAL POWER





BIOMASS

Biomass – organic matter in plants produced through photosynthesis and can be burned directly as a solid fuel or converted into a gas or liquid fuel.

1. Burning wood
2. Agricultural Waste
 - a. Bagasse (sugar cane residue)
 - b. Straw
3. Urban Waste (WTE)
 - a. burning garbage
4. Biofuels
 - a. Biogas – a mixture of 60% methane and 40% carbon dioxide.
 - b. Liquid ethanol- (grain alcohol) – sugar + grain; mix gasoline + ethanol = gasohol which can be burned in conventional gasoline engines (super-unleaded)
 - c. Liquid methanol – wood alcohol

BIOMASS



WOOD



BAGASSE



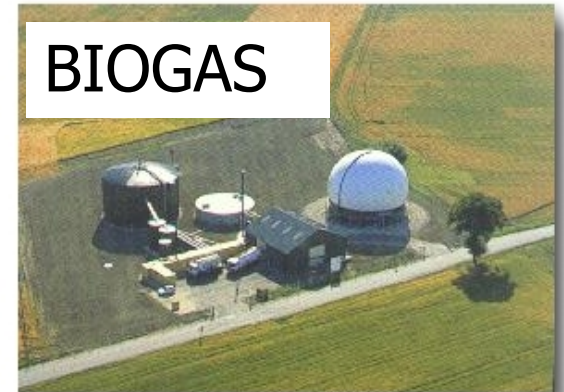
SUGAR CANE

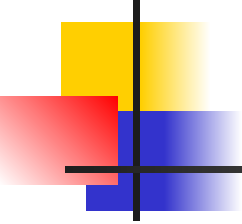


GARBAGE



BIOGAS





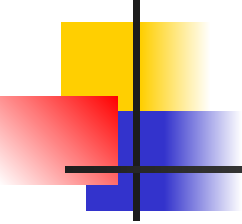
BIOMASS

Advantages

1. potentially renewable resource
 2. less air pollutants released
 3. decrease in use of fossil fuels
 4. moderate-high net energy yield
- reduces biodiversity

Disadvantages

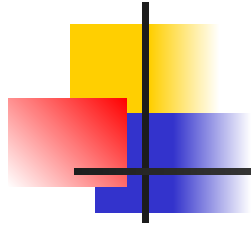
1. removal of trees depletes soil nutrients
2. soil erosion (turbidity)
3. flooding
4. loss of wildlife habitats
5. large land areas needed
6. heavy pesticide/fertilizer use
7. reduces ecological integrity
8. reduces ecological integrity



GEO THERMAL

Geothermal Energy - Heat contained in underground rocks and fluid that can be tapped for energy.

- Extract dry steam, wet steam or hot water and can be used to heat space or water.
- “Potentially renewable resource”
- 22 countries currently use geothermal, it supplies 1% of world energy. In the USA (44% geothermal energy produced worldwide) geothermal electricity is produced mostly in Hawaii, California, Nevada, and Utah.



GEO THERMAL

Advantages

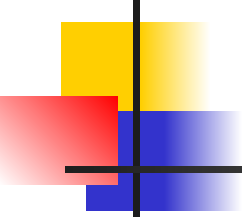
1. Reliable
2. Renewable
3. Moderate Net Energy Yield build plants
4. 96% less CO₂ emitted
5. Competitive Cost

Disadvantages

1. Scarcity of reservoirs
2. Deforestation to
3. Land subsidence
4. Noise, odor

GEO THERMAL





SOLUTIONS FOR SUSTAINABLE ENERGY

- ✓ Improve energy efficiency
- ✓ Increase local availability of renewable energy resources
- ✓ Find transitional resources (natural gas, nuclear)
- ✓ Government must promote R&D for alternative renewable energy resources.
- ✓ Educate the public
- ✓ All energy resources should compete in an open, free-market with NO government control!
- ✓ Government needs to implement constructive subsidies not destructive subsidies to promote change, this will lead to *conservation of resources and less over-consumption.*